

A Real-Time Haptic Simulation Prototype for Liver Lesion Localization Training: System Design, Technical Feasibility Validation, and Formative Expert Evaluation

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ABSTRACT Haptic feedback has potential value in medical training, but the feasibility of integrating physics computation into a real-time haptic loop remains insufficiently explored. To evaluate the system connectivity of a physics engine in an interactive palpation task, this study develops a task-oriented haptic training prototype. Using liver lesion localization as the application scenario, we adopt a Group-Based Corotational Finite Element Method (GB-cFEM) as the physical model, combine it with bimanual tracking and a fingertip-worn interface, and establish a closed loop between large-deformation computation and force response. Hidden lesions are simulated through local modulus enhancement, and the extracted contact-normal reaction force is mapped to haptic output. All runtime tests were conducted on consumer-grade hardware, and a formative expert evaluation was performed to identify engineering boundaries. The results show that the prototype supports system-level operation driven by physics computation and demonstrate the feasibility of this haptic integration architecture and device deployment. Based on these findings, the discussion highlights the challenges that remain before the current physical model can be translated into a clinical-grade training system, including simplified anatomical constraints and the absence of nonlinear material properties, thereby providing design implications and engineering references for future medical palpation systems.

INDEX TERMS Liver palpation, real-time simulation, corotational finite element method, haptic interaction, medical training.

I. INTRODUCTION

Haptic feedback and soft-tissue simulation have long been regarded as important components of medical training systems, but their effectiveness does not depend only on whether haptics is present. It is closely related to task type, evaluation method, and feedback design. Existing reviews show that the value of haptic feedback is often more evident in complex tasks, while training outcomes are still influenced by device degrees of freedom, feedback realism, and experimental design [1]–[3]. For liver lesion localization, operators must compare local stiffness differences under limited visual information. This makes the task a natural testbed for asking whether physics computation can be translated into perceivable haptic cues. Prior work has already shown that liver palpation diagnosis can itself serve as a standalone simulation target, and that liver-related interactive systems are not limited to resection or cutting tasks [4], [5].

From the computational perspective, soft tissues such as the liver exhibit pronounced large deformation and near-incompressible behavior during pressing and traction. Real-time simulation therefore faces a dual constraint of accuracy and efficiency. Continuum FEM offers strong physical interpretability and is well suited to representing local stiffness changes and material-parameter differences, but traditional global solvers are expensive. To address this issue, prior research has explored corotational FEM, cutting FEM, and grouped solving strategies to preserve physical credibility under real-time constraints [6], [7]. At the same time, Projective Dynamics, PBD, and XPBD offer advantages in interactive speed, numerical stability, and implementation convenience [8]–[12]. However, when the goal is to represent lesion stiffness rather than merely visually plausible motion, material interpretability and mechanical consistency remain central.

From the interaction perspective, high-DOF force-

feedback devices can provide richer kinesthetic and contact information, but they usually come with engineering costs such as structural complexity, high price, and restricted workspace [13]. Review studies likewise indicate that medical training systems do not necessarily benefit from blindly pursuing the highest number of degrees of freedom or the highest possible fidelity. What matters more is whether the critical perceptual cues are emphasized for a specific task [1], [2]. Recent work also stresses task-oriented force-profile design rather than reconstructing every haptic dimension by default [14]. This provides theoretical support for the 1-DOF interface used here: if lesion search depends primarily on local differences in normal resistance, then low-complexity hardware may still play a meaningful role in preclinical validation.

Overall, existing studies provide foundations in real-time soft-tissue modeling, liver-related training tasks, and medical haptic-system design, but relatively few works integrate a physically meaningful continuum model, consumer-grade interactive large-deformation updating, and a lightweight haptic loop into a single task-oriented system framework.

Based on this motivation, this paper proposes and implements a task-oriented haptic training prototype to evaluate the system connectivity and engineering feasibility of integrating continuum physics computation into a lightweight haptic loop. The prototype uses liver lesion localization as a concrete test scenario, adopts GB-cFEM as the core physical engine, combines it with Stereo IR-170 bimanual tracking and a haptic interface, simulates hidden lesions via local modulus enhancement, and maps contact-normal reaction force to haptic output. The evaluation in this paper is a formative expert evaluation whose goal is to identify system feasibility and engineering boundaries. The main contributions are as follows:

- 1) A task-oriented GB-cFEM physical-haptic closed-loop system prototype is proposed and implemented. Contact-normal force is extracted from a continuum physics engine and translated into force output, demonstrating the system connectivity of this computational architecture for palpation interaction and enabling real-time deployment on compact consumer-grade hardware.
- 2) A haptic mapping and exploration mechanism for lesion localization is designed. By combining local stiffness encoding with a fingertip-worn haptic interface, the system supports real-time extraction and feedback of hidden-lesion mechanical features and demonstrates the application potential of local hardness differences as palpation cues.
- 3) A formative evaluation involving a surgical expert is conducted. Through walkthrough-based assessment, the study evaluates the prototype's ability to transform FEM-based stiffness into recognizable mechanical cues and identifies the engineering boundaries and practical constraints that remain before translation to a clinical-grade trainer.

II. RELATED WORK

Interactive simulation systems for surgical training and palpation-skill acquisition typically require engineering tradeoffs among visual rendering rate, haptic update frequency, and physical-modeling complexity. Reviews in this area note that virtual-reality and haptic-feedback systems for medical training are limited not only by tissue-mechanics modeling and real-time rendering, but also by task definition, evaluation methodology, and the maturity of educational evidence [1]–[3]. For the liver palpation training task studied here, the most relevant research threads concern real-time soft-tissue modeling, liver-related training tasks, and low-dimensional haptic rendering.

A. REAL-TIME SOFT-TISSUE MODELING FOR SURGICAL SIMULATION

In soft-tissue modeling for surgical simulation, continuum FEM is a common and important family of methods because it naturally represents material parameters, boundary conditions, and volumetric mesh structure, although it becomes computationally expensive under large deformation, near-incompressibility, contact, and cutting. Corotational FEM separates large rotation from strain computation and achieves a favorable balance between efficiency and deformation quality, and has been used in real-time surgical settings such as needle insertion [6]. Most directly relevant to this work is GB-cFEM, which combines grouped solving, domain decomposition, and offline precomputation to support large tetrahedral models, anisotropic materials, and near-incompressible cases on multicore CPUs [7]. This makes it especially suitable as the core of the liver large-deformation simulation used here.

Beyond traditional FEM, Projective Dynamics and its extensions for hyperelastic materials also provide a balance between stability and real-time performance [8], [9], while soft-tissue models based on constraint energies further improve physical consistency in tool-tissue interaction [15]. By comparison, PBD and XPBD are often used as engineering baselines in real-time systems because of their implementation simplicity and numerical robustness [10]–[12]. However, when anisotropy, near-incompressible volume preservation, and organ-level physical interpretability are required, FEM generally remains the more appropriate primary model.

B. LIVER-RELATED SIMULATION SYSTEMS AND TRAINING TASKS

In interactive simulation for the liver and other abdominal organs, previous studies span both organ deformation and cutting workflows and palpation-based lesion exploration and training. Prior work has reported a liver-parenchymal transection system with haptic feedback [5], while another study presented a haptic simulation system for liver diagnosis through palpation [4]. These studies indicate that liver-related training is not limited to cutting. Palpation, localization, and stiffness comparison also have independent training value. Accordingly, the present work does not attempt to reproduce a full hepatectomy workflow, but instead focuses on the more

clearly defined and more readily evaluable subtask of hidden-lesion palpation and localization.

C. HAPTIC DEVICE DEGREES OF FREEDOM AND LOW-DIMENSIONAL HAPTIC RENDERING

In medical haptic rendering, device degrees of freedom, force-mapping strategy, and system complexity are tightly coupled. Existing reviews describe both multi-DOF force-feedback interfaces and lightweight low-cost solutions [1]–[3]. For example, Agboh *et al.* proposed a six-DOF laparoscopic training interface, demonstrating the feasibility of high-DOF haptic reconstruction [13]. At the same time, growing evidence suggests that perceptual cues should be selected according to the task goal rather than assuming that full-dimensional, high-fidelity reconstruction is always necessary. Beyond traditional desktop devices, recent work has also proposed a lightweight fingernail-worn haptic device that provides force and vibration cues while minimizing obstruction of the natural fingerpad contact area, offering a new hardware path for low-burden bare-hand interaction [16].

Along this line of reducing haptic dimensionality or emphasizing specific cues, reality-based force profiles have been used to simplify kinesthetic rendering in surgical simulation [14]. These studies suggest that low-DOF haptic output is not merely a simplified substitute, but can also be understood as a task-oriented design choice for highlighting perceptual cues such as normal resistance and stiffness contrast.

D. POSITION OF THIS WORK

In summary, existing studies have established important foundations in real-time soft-tissue modeling, liver-related training tasks, and medical haptic rendering. However, there is still room to explore system-level integration of a physics engine and a lightweight haptic loop under consumer-grade hardware, together with system-level diagnosis. Based on this gap, this paper develops a task-oriented haptic training prototype using liver lesion localization as the test scenario and focuses on validating the system integration and usability of a physics-computation-driven interaction architecture. The evaluation in this paper is a formative expert evaluation whose goal is to identify system feasibility and engineering boundaries rather than to demonstrate training efficacy.

III. SYSTEM DESIGN AND IMPLEMENTATION

A. OVERALL SYSTEM ARCHITECTURE

The system consists of four layers: a real-time physics simulation layer, an interaction input layer, a haptic rendering layer, and a task configuration layer. The real-time physics simulation layer uses GB-cFEM to solve the liver tetrahedral mesh in real time. The interaction input layer uses a Stereo IR-170 (Ultraleap, UK) to capture bimanual hand pose, with the five fingertips of the right hand corresponding to five virtual proxy spheres. The haptic rendering layer sends the contact-normal reaction force to a 1-DOF device through a serial protocol. The task configuration layer organizes blind

lesion-localization experiments by switching among internal hard-region positions.

B. REAL-TIME SIMULATION CORE

GB-cFEM is adopted as the real-time simulation core. The liver tetrahedral mesh is first divided into multiple local groups, each of which is solved independently in a corotational coordinate system for local deformation. During the off-line stage, system matrices are prefactorized. At runtime, the main operations are rotation extraction, right-hand-side construction, and iterative computation. Continuity across groups is maintained through coupling constraints. Compared with a full global implicit FEM, this design avoids reassembling and refactorizing a large linear system every frame. Compared with XPBD, it preserves clearer material-parameter meaning and is therefore better suited to representing stiffness differences between normal tissue and lesion regions.

C. BIMANUAL INPUT AND VIRTUAL PALPATION INTERACTION

The system uses the Stereo IR-170 for bimanual tracking. Fingertips are mapped to capsule-shaped proxy bodies used for local pressing, lesion localization, and surgical-field exposure. The right hand is primarily responsible for palpation and comparison, while the left hand is used mainly to push and retract tissue so as to create a local exposure window for the right hand to enter. To reduce tracking noise and the numerical shock caused by instantaneous penetration, the proxy bodies are not set equal to the raw sensor positions. Instead, they are connected to the device-side positions through damped virtual couplings.

D. 1-DOF HAPTIC RENDERING STRATEGY

The contact-normal reaction force is used as the main input to the 1-DOF device. The fingertip-worn haptic device used here follows the lightweight fingernail haptic design proposed by the authors previously. In this design, a motor drives a fine wire to apply normal force on the fingerpad side, while the lightweight actuation mechanism is integrated on the fingernail side to minimize occlusion of the fingerpad and reduce interference with natural touching behavior [16]. Because optical hand tracking has difficulty accurately distinguishing little-finger motion and can generate erroneous feedback, the experimental haptic device was implemented on the four fingers of the right hand except the little finger. On this basis, the device is further integrated into the real-time GB-cFEM simulation loop to convey local stiffness differences in the lesion-search task. The system extracts the contact-force vector $\mathbf{F}_{\text{contact}}$ and the surface normal $\mathbf{n}$ from the simulation core and computes the normal reaction-force magnitude delivered to the haptic hardware as

$$F_n = |\mathbf{F}_{\text{contact}} \cdot \mathbf{n}|. \quad (1)$$

In the deployed system, the actual update rate of the haptic control loop is 1000 Hz, and the physics engine transmits

force commands to the haptic hardware through a serial link at 2,000,000 bps. Under this mechanism, users perform lesion localization by perceiving differences in normal resistance while pressing different regions. The hardware connections and wearable device used in the experiment are shown in Fig. 1.

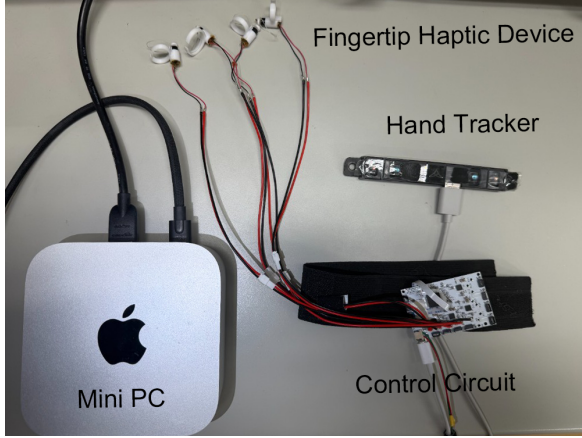


FIGURE 1. Hardware setup used in the experiment. The lower-right part shows the wrist-mounted fixture and driving circuit, while the upper part shows the fingertip haptic actuator. Together with Stereo IR-170 tracking, the system provides hand tracking and normal-force cues.

E. LIVER MODEL AND LESION CONFIGURATION

Under the default configuration used in the physician experience session, the main system parameters are listed in Table 1. These parameters are intended to support a stable and perceptible interactive demonstration and are grounded in the deformation and volume behavior demonstrated by the underlying technical validation. In particular, the “tetrahedral volume constraint correction” in Table 1 is not an inherent analytic property of the main GB-cFEM solver. It is an additional stabilization step enabled in the current system. Intuitively, when strong local pressing causes certain tetrahedra to become too flattened, approach inversion, or deviate noticeably from their rest volume, the system applies a small positional correction to pull them back toward the rest volume. This helps prevent local collapse, abnormal shrinkage, and subsequent contact jitter. The evaluation of the system’s “physical consistency” in this paper therefore comes both from algorithm-level validation data and from the task performance and subjective feedback of a clinical expert during actual interaction.

To support blind localization experiments, the system pre-sets three switchable local hard-region positions inside the liver. The hard regions are not indicated by visible geometric shape, but are hidden inside the tissue through embedded Young’s-modulus enhancement. This forces participants to rely only on local deformation and haptic differences to infer lesion location, which is closer to the nature of palpation. In addition to lesion configuration, fixed points are defined in three anatomical regions of the liver model to suppress non-physical global drift and approximate basic attachment relations: the Porta Hepatis, the Groove for the Inferior Vena Cava

TABLE 1. Liver Model and Lesion Parameters Under the Default Physician-Experience Configuration

Parameter	Value	Description
Geometry source	NIH 3D (3DPX-021007)	Open liver anatomy model
Meshing parameter	TetGen -pq20a0.00083	Controls tetrahedral mesh quality and density
Base Young’s modulus	7000 Pa	Baseline stiffness of normal liver tissue
Tumor-region Young’s modulus	30000 Pa	Local stiffness enhancement in hard regions
Poisson ratio	0.49	Near-incompressible setting
Number of groups	288	Local grouped-solving configuration
Time step	0.008 s	Real-time simulation step
Tetrahedral volume constraint correction	Enabled (2 iterations, correction = 0.15)	Pulls tetrahedra back toward their rest volume in each step to suppress local collapse, inversion, and abnormal shrinkage under deep pressing

(IVC groove), and the Bare Area, as shown in Fig. 3. This multi-point, surface-based anchoring strategy approximates the stable connection between the liver and major vascular and diaphragmatic attachment regions while preserving local deformability.

Beyond these anatomical anchors, a static liver-shaped abdominal support boundary is also introduced to approximate the external constraints imposed by surrounding abdominal tissue. Specifically, the system extracts the contact surface from the outer triangular mesh of the liver in the rest pose and offsets it outward along the surface normal of each facet by a small clearance proportional to the bounding-box diagonal, thereby constructing an envelope shell that follows the liver geometry while being slightly expanded. It should be emphasized that this shell is not designed as a fully closed abdominal box. Instead, to match the operating conditions of open surgery, a local opening is deliberately retained along a selected direction and serves as an exposure window created after incision and retraction, allowing fingers to enter from the exposed side for palpation.

At runtime, contact between surface vertices and this static shell is computed through local adjacent-triangle search and closest-point projection. When a vertex crosses the allowable clearance toward the outside of the shell, the system applies a push-back correction along the shell’s outward normal, combined with tangential damping and velocity update to suppress oscillation. Accordingly, the “abdominal collision” in this paper should be understood more precisely as a liver-shaped support envelope generated from the rest-pose geometry rather than a high-fidelity anatomical reconstruction of the abdominal wall, diaphragm, and surrounding organs. This is also one of the main limitations of the current model. Although it provides more plausible spatial support than a simple planar wall, it does not yet model the anterior falciform

ligament or authentic abdominal contact relations individually. The runtime interface and the division of left-hand exposure and right-hand palpation are shown in Fig. 2.

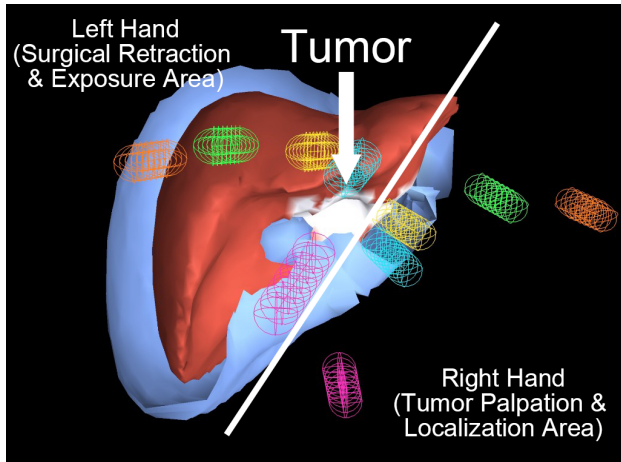


FIGURE 2. Runtime screenshot of the system. The red region is the liver model, and the outer light-blue shell is the liver-shaped collision cavity used to provide spatial constraint. One side is retained as an open region to approximate the local exposure window in open surgery. The colored wireframe capsules denote fingertip proxy bodies. The left-side region corresponds to the left hand, used for pushing and retracting tissue to maintain exposure, while the right-side region corresponds to the right hand, used for lesion palpation, pressing-based comparison, and localization search.

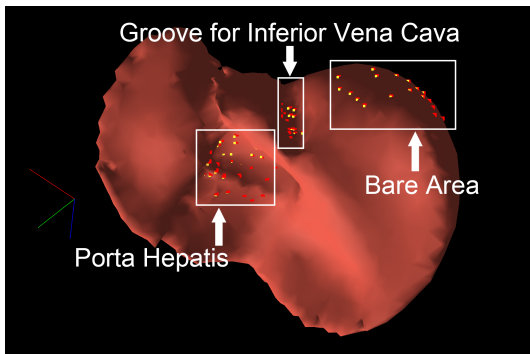


FIGURE 3. Anatomical fixation regions and fixed-node distribution in the liver model. The dotted markers denote node clusters with fixed-point constraints, corresponding to the Porta Hepatis, the Groove for the Inferior Vena Cava, and the Bare Area. These constraints approximate major anatomical attachments and suppress nonphysical rigid-body drift.

IV. EXPERIMENTAL DESIGN AND EVALUATION METRICS

A. APPLICATION-LEVEL TECHNICAL VALIDATION BASED ON A SPECIFIC LIVER MODEL

Although the systematic benchmark of the underlying pure algorithmic GB-cFEM framework, such as idealized tests on basic geometries, has already been published in [7], applying it to an actual medical model within a haptic loop imposes new requirements on real-time performance and physical stability because of increased model complexity. Therefore, instead of relying on the generic 3D models used in [7], this paper re-runs validation on the liver anatomical model used by the present system in order to show that the computational

framework can still serve as the real-time basis of the current interactive system:

- 1) Large-deformation accuracy: displacement response of the proposed method and XPBD is compared under unified material parameters and loading conditions, using VegaFEM as the high-accuracy reference.
- 2) Near-incompressible volume preservation: under large dragging with different Poisson-ratio settings, the maximum and mean deviations of the volume ratio V/V_0 are measured.
- 3) Anisotropic response validation: force-displacement slopes under loading in different directions are compared to examine whether the system can stably convey direction-dependent mechanical differences.
- 4) Real-time performance and scalability: the physics-solver frame rate is measured at different tetrahedral scales to assess interactivity under standard CPU conditions.

In this paper, “XPBD Fast” refers to the typical real-time configuration used in the corresponding script (`subSteps = 5`, `maxIterations = 1`), whereas “XPBD Ref” refers to a higher-accuracy reference configuration under the same mesh and the same fixation strategy but with XPBD substeps increased substantially to 50 in order to observe its error and cost near convergence.

Both XPBD configurations are reported to avoid the bias of comparing only one setting. If only the low-substep configuration were reported, XPBD would be placed at an obvious accuracy disadvantage because of its emphasis on real time. If only the high-substep configuration were reported, its interactive cost would be obscured. Reporting both “Fast” and “Ref” makes the typical tradeoff of XPBD between real-time performance and stiffness/deformation accuracy easier to interpret fairly and clarifies the relative advantage of the method used in this paper.

Unless otherwise stated, all system runtime tests, real-time demonstrations, and the formative expert evaluation in this paper were conducted on the same Mac mini (Apple, USA) with an Apple M4 Pro (12-core CPU), 24 GB memory, and macOS 26.3.1. This is emphasized because the goal of the paper is not only theoretical real-time performance, but also actual deployability on compact consumer-grade hardware. As a supplementary whole-machine observation, when the system ran continuously for 5 minutes on the same device, the wall power averaged about 52 W, compared with about 12 W at idle, indicating that the additional whole-machine power required by the real-time FEM interaction pipeline is on the order of only several tens of watts rather than a high-power workstation load.

B. FORMATIVE EXPERT EVALUATION AND OBSERVATION OF LESION-SEARCH BEHAVIOR

One surgeon participated in the system evaluation. The participant profile is listed in Table 2. The study protocol was approved by the ethics committee of the Institute of Science

Tokyo (approval number: 2023362). The participating surgeon was informed about the study before the experiment and signed an informed-consent form.

This evaluation was a formative expert evaluation. Its goal was to identify system feasibility and engineering boundaries rather than to prove training effectiveness. The evaluation setup is shown in Fig. 4. The expert first watched an approximately 5-minute system demonstration and then completed, in sequence, free exploration, a blind lesion-localization observation, and a usability evaluation task.

In the lesion-localization task, to reduce spatial-memory effects, different unknown lesion positions were used under the two conditions of haptics disabled and haptics enabled. The expert's search behavior and strategy changes under the two conditions were recorded for qualitative analysis of how the local stiffness cues provided by the system supported palpation judgment.

TABLE 2. Profile of the Surgical Expert Participating in the Formative Evaluation

Attribute	Information
Professional title	Intermediate
Clinical experience	7 years
Specialty	Surgery
Surgery-related experience	2 years

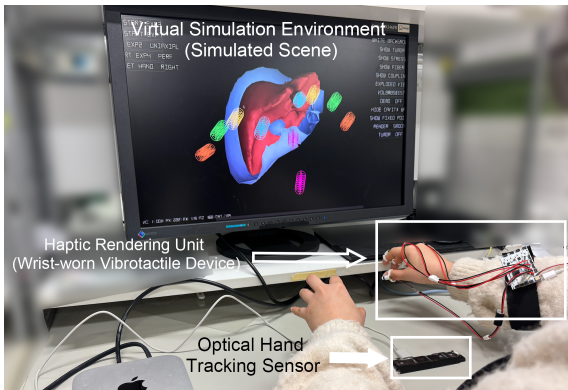


FIGURE 4. Formative expert evaluation setup. The participant performed virtual palpation through hand motion while receiving 1-DOF force feedback, and the screen displayed real-time liver deformation and proxy-body positions.

The task design and recorded contents of the formative expert evaluation are listed in Table 3.

C. METRIC CONSTRUCTION FOR THE FORMATIVE EVALUATION

The formative evaluation in this paper records three kinds of indicators: behavioral indicators (changes in lesion-search strategy), subjective evaluation (verbal feedback during free exploration and semi-structured interview), and rating-scale scores (evaluation of soft-tissue information expression and system performance). The assessment emphasizes qualitative

TABLE 3. Task Design and Recorded Contents of the Formative Expert Evaluation

Task	Goal	Condition	Recorded content
Task A: Free exploration	Observe interaction fidelity and naturalness	Freely touch the liver while verbalizing thoughts	Subjective evaluation and interaction behavior
Task B: Blind lesion localization	Observe search-behavior changes across two trials	Trial 1: haptics off; Trial 2: haptics on with a different lesion position	Operational strategy and verbal feedback
Task C: System usability feedback	Evaluate engineering boundaries and physical consistency	Press different regions and compare resistance differences	Semi-structured interview and Likert-scale ratings

observation and system-mechanism analysis in order to provide diagnostic and design guidance for the next generation of the system.

V. RESULTS

A. TECHNICAL VALIDATION RESULTS

1) Large-Deformation Accuracy and Real-Time Performance
When the test was re-run on the real liver model used in this paper, with VegaFEM taken as the reference, Fig. 5 and Table 4 show that the kernel used by the present system achieved an average displacement error of about 17.9% under three loading strengths, whereas the error of XPBD Fast exceeded 50%. More importantly, for the liver-lesion palpation scenario considered here, which contains tens of thousands of tetrahedra, the method still achieved about 83 FPS at a computational cost of approximately 12 ms per frame, satisfying the real-time interaction requirement of the system. Although XPBD reached higher FPS at low substeps, its displacement response was overly soft, whereas the higher-accuracy configuration sacrificed real-time capability. Considering that the system test was carried out on a Mac mini M4 Pro and that the average wall power of the full machine was approximately 52 W during continuous 5-minute operation (versus about 12 W at idle), this result at least indicates that the FEM-driven pipeline is not limited to heavy computing platforms and can support actual interactive operation with only several tens of watts of additional whole-machine power.

TABLE 4. Comparison of Large-Deformation Accuracy and Real-Time Performance

Metric	Proposed method	XPBD Fast	XPBD Ref	VegaFEM
Mean displacement error	17.9%	50.8%	43.6%	0% (reference)
Mean runtime	12.0 ms	7.9 ms	59.2 ms	—
Frame rate	83.3 FPS	126.6 FPS	16.9 FPS	—

Taken together, this result provides supporting evidence for the following judgment: the simulation kernel relied upon by the present system does not simply trade accuracy for

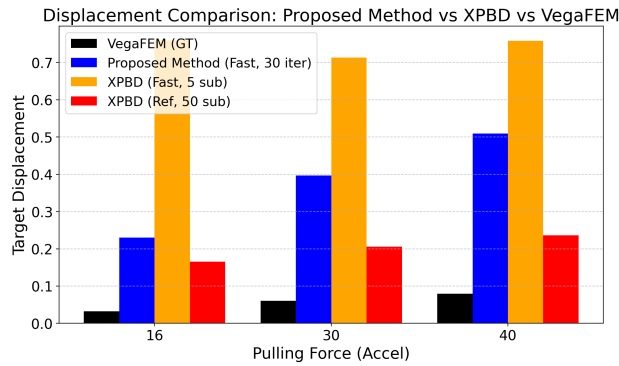


FIGURE 5. Comparison of large-deformation displacement. Relative to the VegaFEM reference, the proposed method exhibits more stable displacement response than both XPBD Fast and XPBD Ref under all three loading strengths.

speed, but maintains a more stable stiffness representation than XPBD within an interactive operating range.

2) Near-Incompressible Volume Preservation

Specialized validation under the typical near-incompressible setting of a high-water-content soft tissue such as the liver shows that, as presented in Fig. 6 and Table 5, the volume stability of the kernel used here is overall better than that of XPBD and close to the VegaFEM reference. In particular, under the $\nu = 0.47$ condition, the mean volume deviation of the tested liver model was 0.92%, and the maximum deviation was 1.35%, both better than XPBD.

TABLE 5. Volume-Deviation Comparison Under Near-Incompressible Conditions

Method	Poisson ratio	Maximum volume deviation	Mean volume deviation
Proposed method	0.28	1.88%	1.31%
Proposed method	0.47	1.35%	0.92%
XPBD	0.28	7.27%	1.84%
XPBD	0.47	1.59%	1.29%
VegaFEM	0.28	1.94%	1.28%
VegaFEM	0.47	1.47%	0.80%

This indicates that, under the current protocol, the proposed method is less likely to produce the appearance of abnormal shrinkage or collapse under large pressing and dragging, thus providing a more consistent geometric basis for subsequent haptic mapping.

3) Real-Time Performance and Scalability

Following the performance framework of [7], we further evaluated the method at different mesh scales. As shown in Fig. 7 and Table 6, the simulation kernel reached 219.24 FPS at approximately 7,790 tetrahedra and still maintained about 20.2 FPS at approximately 35,486 tetrahedra. By contrast, VegaFEM achieved only 5.52 FPS at about 18,245 tetrahe-

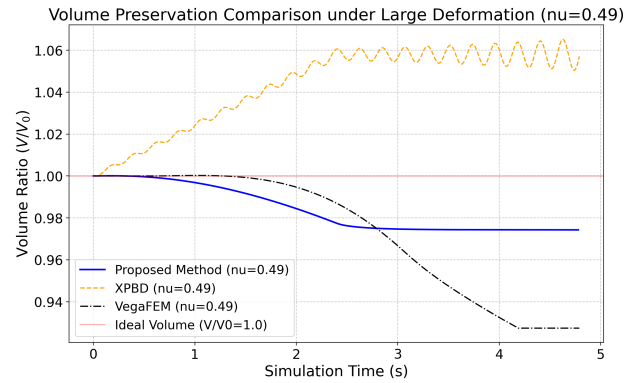


FIGURE 6. Comparison of volume preservation under near-incompressible conditions. The proposed method maintains small volume deviation under both Poisson-ratio settings and is overall close to the VegaFEM reference.

dra, and the high-accuracy XPBD reference configuration achieved 16.27 FPS at about 20,061 tetrahedra.

TABLE 6. Comparison of Real-Time Performance Across Mesh Scales

Method	Mesh scale	FPS	Note
Proposed method	7,790 tets	219.24	Above real-time interaction demand
Proposed method	35,486 tets	20.2	Maintains basic interactivity
XPBD Ref	20,061 tets	16.27	High accuracy comes at high cost
VegaFEM	18,245 tets	5.52	Difficult to use interactively

Although the mesh scales used by different methods are not matched point by point, this result is still sufficient to indicate that the GB-cFEM route used by the current system can support medical demonstration and interactive tasks on a consumer-grade CPU without relying on heavier high-end computing conditions.

B. RESULTS OF THE FORMATIVE EXPERT EVALUATION AND SYSTEM-USABILITY FEEDBACK

1) Interaction Fidelity During Free Exploration

During the free-touch task, the expert participant first pressed different parts of the liver surface with multiple fingers, then attempted to lift the thin edge of the liver, and further explored the dynamic change of force feedback by varying the pressing intensity. Based on comparative palpation between the preset tumor region and normal tissue, the expert identified the engineering limitations of the system from both anatomical and physical perspectives.

The first issue concerned distortion of material behavior. The expert pointed out that the current liver soft-tissue model was overly elastic, with a feel closer to homogeneous soft gel than to fibrous biological tissue. A real liver exhibits nonlinear stiffness and is covered by a capsule, producing an initial resistance followed by constrained compression at

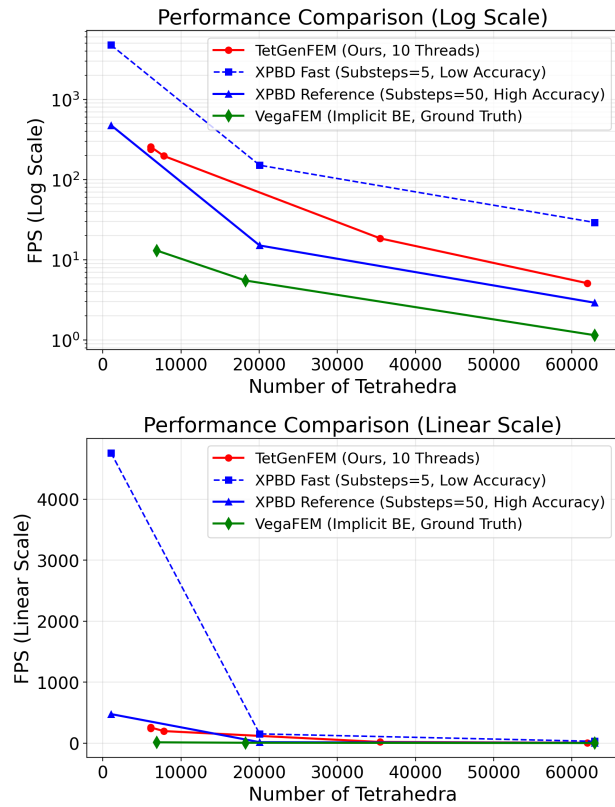


FIGURE 7. Overall performance trend across mesh scales. The proposed method maintains clearly better real-time performance than VegaFEM over a broad tetrahedral range and still retains basic interactivity on larger-scale models.

deeper indentation. By contrast, the present model deforms too easily under compression.

The second issue concerned local perceptual differences caused by simplified anatomical constraints. The expert observed that when the liver was pushed from the side, the model exhibited global translation similar to a floating object rather than rotation or local deformation constrained by anterior ligament traction. In fact, the boundary conditions in the physical model were simplified through an equivalent representation. Instead of adopting computationally expensive one-dimensional ligament models, fixed points were directly assigned to the Porta Hepatis (vascular root anchor), the Bare Area (a broad superior adhesion region), and the Groove for the Inferior Vena Cava (major venous support). Specifically, fixing the Porta Hepatis is equivalent to locking the structural pivot at the bottom of the liver; fixing the whole surface of the Bare Area computationally covers the tensile limiting effect that would otherwise be imposed by its marginal ligaments; and fixing points along the IVC groove is equivalent to providing a vertical load-bearing support on the posterior side. This surface-based macroscopic fixation strategy already captures the main traction-limiting roles of the supporting ligaments at the mechanical level and is used to support stable interaction under large deformation. The expert's feedback suggests that, although this physical equivalent simplifica-

tion is mechanically reasonable at a macroscopic level, the absence of anterior traction constraints can still produce a mismatch between local physical feedback and actual clinical logic during specific lateral pushing maneuvers.

The final issue concerned deviation from the clinical interaction paradigm. The expert noted that bare-hand palpation is part of clinical diagnosis, but in surgical contexts the main interaction medium is a surgical instrument. Direct hand-tracking interaction sacrifices part of the task validity of surgical simulation and makes the experience closer to physical examination than to operative practice.

These comments define the current use scenario of the system as a targeted training prototype focused on expressing lesion-related haptic characteristics and clarify the mechanical and boundary conditions that still need to be added before approaching clinical-grade surgical simulation.

2) Behavioral Observation in the Lesion-Localization Task

In the initial exploration with haptic feedback disabled, the expert relied mainly on visual deformation for judgment. Observation showed that the expert first attempted light surface pressing and, after failing to obtain useful stiffness cues, gradually shifted to larger pulling motions and deeper indentation.

In the localization test with haptic feedback enabled, where the lesion was placed in a new unknown region, the expert's search behavior changed to local comparative light pressing. Verbal feedback indicated that the local differences in normal resistance provided by haptic feedback supported discrimination of the hard region. This strategy change suggests that the FEM-driven stiffness-mapping logic can provide perceptible physical cues for the lesion-search task.

The expert's operational feedback reveals the boundary between this engineering prototype and a clinical trainer. These qualitative observations play a diagnostic engineering role by providing concrete experiential guidance for future improvements in medical-simulation fidelity.

3) Diagnosis of Engineering Constraints and Evaluation of Force-Rendering Quality

In Task C, the expert actively performed continuous sliding exploration and deep pressing on the liver surface while cross-checking fingertip force sensation against the visual deformation shown on the screen. Based on these dynamic palpation behaviors, the expert identified a core engineering bottleneck in the system connectivity: the hardware limitation in feedback directionality. The current 1-DOF output can convey only one-directional normal resistance, while lacking tangential force and more complex reaction-force distribution. As a result, it cannot reproduce the enveloping squeeze sensation that a real finger experiences when pressed into tissue and surrounded by the material. This reflects an inherent limitation of low-dimensional hardware. The corresponding Likert-scale ratings are listed in Table 7.

According to the Likert-scale scores in Table 7, the system received positive ratings in "architectural feasibility" and "soft-tissue mechanical expression" (4/5), while the score

TABLE 7. Likert-Scale Ratings of System Usability and Force-Rendering Quality

Evaluation dimension	Score (out of 5)	Interpretation
System feasibility and potential	4/5	The “continuum mechanics + haptics” system architecture was considered feasible
Soft-tissue mechanical expression	4/5	The stiffness settings of the physical engine could be effectively mapped into haptic cues
Force-feedback realism	3/5	The physical limitation of single-DOF hardware dimensionality and refresh behavior was revealed

for “force-feedback realism” was only moderate (3/5). This result is not intended to prove the absolute superiority of the system. Rather, it objectively confirms that the data flow of the system has been established and clarifies the engineering difficulties that the next-generation system must overcome in hardware frequency response and feedback dimensionality.

VI. DISCUSSION

A. FROM ALGORITHM VALIDATION TO SYSTEM-PROTOTYPE EVALUATION

This point needs to be stated clearly. The prior GB-cFEM work addressed real-time large-deformation physical solving, and the previous haptic-device work addressed lightweight fingertip output. The new work of this paper lies in the task-oriented integration of a medical palpation scenario, the normal-force-mapping closed loop, deployment on consumer-grade hardware, and formative expert diagnosis.

This paper focuses on validating the feasibility of a palpation interaction system driven by a physical kernel. The application-level technical validation provides supporting evidence that the simulation kernel can support real-time stiffness expression on a medical model, while the formative evaluation identifies the boundaries of the model as an interactive prototype through walkthrough-based diagnosis. The issues highlighted in the evaluation, including simplified anatomical constraints, the absence of material nonlinearity, and the directional limitation of 1-DOF output, are not merely generic engineering limitations. They directly affect the judgment logic of the core task of lesion localization.

First, the absence of material nonlinearity blurs the perception of lesion depth and stiffness. The current model deforms too easily under compression and lacks the progressive compressive resistance and capsule-related limitation of real tissue during deep pressing. In the localization task, this linear response makes it difficult for the operator to probe deep hard nodules through changes in pressing resistance and therefore weakens palpation-based discrimination of deep lesions.

Second, simplified anatomical constraints such as anterior traction affect recognition of marginal lesions. Although the current anchoring strategy guarantees macroscopic stability, the organ can still undergo global translation rather than local

stretch during lateral pushing. When a lesion is located near the liver edge, the operator may feel reaction force dominated by organ displacement rather than by the local hardness abnormality of the lesion itself, which interferes with search judgment near the organ boundary.

Third, the normal-force limitation of 1-DOF feedback reduces the ability to perceive lesion boundaries. Because tangential force and lateral squeezing cues are absent, the operator cannot perceive the resistance-gradient change or the spatial sense of envelopment while the finger slides across the edge of a hard region. This increases the difficulty of outlining a lesion in three dimensions and forces the operator to rely on dense vertical pressing to infer the boundary.

These mechanisms, which are difficult to expose through numerical tests alone, reveal how engineering simplifications can change the behavior pattern of lesion localization. The construction of the present system prototype therefore not only verifies the connectivity of the data flow, but also provides concrete targets for future system evolution through diagnostic analysis.

B. APPLICATION FOCUS AND FUTURE EVOLUTION OF THE SYSTEM

For the test scenario of liver lesion localization, the current system expresses lesion characteristics through normal-force-based stiffness feedback. At the same time, what is validated here is not an isolated implementation for one specific liver model, but a transferable way of organizing the system. As long as the target task also depends on local stiffness differences, contact-reaction cues, and real-time updatable soft-tissue deformation, the architecture of “GB-cFEM physical kernel + local property encoding + haptic mapping” may be extended to other soft-tissue palpation or lesion-search scenarios. Of course, such extension is not unconditional and still requires organ-specific remodeling of geometry, boundary constraints, material parameters, and interaction paradigm. To further improve the immersion of surgical simulation as a whole, future evolution may proceed in the following directions: first, introducing more complex material models with capsule effects and nonlinear compressive behavior; second, adding more realistic ligament and abdominal support constraints to strengthen organ anchoring; third, extending the interaction paradigm smoothly from bare-hand palpation to instrument-dominant workflows; and fourth, while preserving the lightweight advantage of the 1-DOF solution, further improving tangential compensation and continuity in feedback control.

C. FUTURE RESEARCH DIRECTIONS

Future work may further examine task validity and training effectiveness under a larger sample, but the scope of the present paper is limited to system-prototype design and technical feasibility validation. The next stage of engineering iteration will prioritize the following directions: first, enriching ligament and abdominal-boundary modeling to improve the macroscopic physical support of the organ; second, exploring

surgical-instrument proxy interaction to correct the current interaction paradigm; and finally, optimizing the synchronization mechanism between the underlying physics engine and the haptic serial loop to improve the continuity of force rendering.

VII. CONCLUSION

This paper proposes and implements a task-oriented haptic training prototype to verify the system connectivity and engineering feasibility of integrating a GB-cFEM physical engine with a haptic interface. By combining GB-cFEM-based real-time soft-tissue simulation with haptic mapping and using liver lesion localization as a concrete test scenario, the study realizes the transformation of a physical simulation kernel into an interactive feedback engine.

The technical validation provides application-level evidence regarding robustness, real-time performance, and computational economy on a medical model, showing that the system can run stably on ordinary consumer-grade hardware. However, these comparative results should still be understood as comparative validation for a system prototype rather than as a fully standardized cross-method benchmark. The evaluation in this paper is a formative expert evaluation whose goal is to identify system feasibility and engineering boundaries rather than to prove training effectiveness. The assessment by a single surgical expert identifies the engineering challenges and boundaries that must be addressed before the prototype can move toward a high-fidelity clinical trainer, including richer nonlinear material properties, more complete anatomical constraints, and expanded feedback dimensionality.

Overall, this paper validates the feasibility of a “continuum physics computation + task-oriented haptic feedback” architecture. Through system-level diagnosis in a formative evaluation, the study clarifies the current design limitations and engineering boundaries of the system and provides engineering evidence and design insight for exploring lightweight palpation solutions and building the next generation of medical physical-interaction systems.

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